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60-0000-L-PRO-0001

DOCUMENT TITLE:

**PROCEDURE FOR LINE NUMBERING
AND LINE LIST GENERATION**

ДОКУМЕНТ №

60-0000-L-PRO-0001

НАИМЕНОВАНИЕ

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TENGIZCHEVROIL

PFD Project No. 571500

ТЕНГИЗШЕВРОЙЛ

PFD Проект № 571500

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1. SCOPE

The purpose of this document is to define and maintain a consistent approach to Line Numbering on IP & ID's and Line Lists, for the TCO Second Generation, and Gas Injection Development, Projects.

2. ABBREVIATIONS.

PFD	Parsons Fluor Daniel
IP&ID	Intelligent Piping & Instrumentation Diagram
TCO	Tengizchevroil
SGP	Second Generation Project
SGI	Sour Gas Injection
PME	Piping Materials Engineering (Group)
PE	Process Engineering (Group)
WBS	Work Breakdown Structure
NDT	Non Destructive Testing
PWHT	Post Weld Heat Treatment

3. INTRODUCTION.

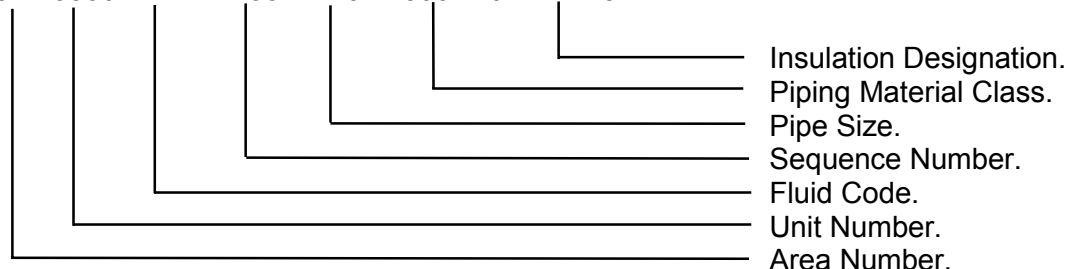
The Project Piping Line Numbering Procedure provides the convention, mechanics, and responsibilities for designating and numbering of process and utility lines. This procedure includes a description of line designator components, the method for assigning line sequence numbers, and examples of the numbering system.

4. DESCRIPTION OF LINE DESIGNATOR.

The Line Designator or Number is the combination of information that fully specifies the pipeline in accordance with the project Work Breakdown Structure (WBS).

A complete line designator is shown in the following example: -

62 - 0300 - PHC - 1001 - 10" - 300H2O - HCW5

**4.1 Area and Unit Code.**

The Area and unit code numbers are derived from document no. 60-0000-A-SKH-0001 - "Work Breakdown Structure " (WBS).

4.2 Fluid Code

The Fluid Codes are listed in the “Fluid Code Selection Table” of Document No. 60-0000-L-SPE-0003 – Specification for Piping and Material Classes.

4.3 Sequence Number

Sequence Numbers are four-digit numbers. Unique sequence numbers will be allocated for each fluid code and will be assigned by the Piping Materials Engineering (PME) group. The line Number will maintain the same sequence number across all areas.

4.4 Pipe Size.

Nominal pipe diameter is indicated in inches.

4.5 Piping Material Class

The Piping Material Classes are listed in the “Piping Line Class Selection Tables” of Document No. 60-0000-L-SPE-0003 – Specification for Piping and Material Classes.

4.6 Insulation Designation.

The piping systems insulation designation codes are listed in the “Piping Systems Insulation Selection Table” of Document No. 60-0000-L-SPE-0003 - Specification for Piping and Material Classes.

5. RESPONSIBILITIES**5.1 Lead Piping Materials Engineer.**

- (a) Ensure that the correct line numbering procedure is followed.
- (b) Issue of Line List or Summary.
- (c) Provide Hydrotest Pressure and Medium, NDT %, PWHT and Paint for entering into IP & ID database.

5.2 Piping Materials Engineering Group.

- (a) Control of Line Sequence numbering on the project.

5.3 IP & ID Production Group.

- (a) Entering of the Line Sequence number and all associated data into IP&ID database.
- (b) Line List reports.
- (c) Propagation of information to the database.

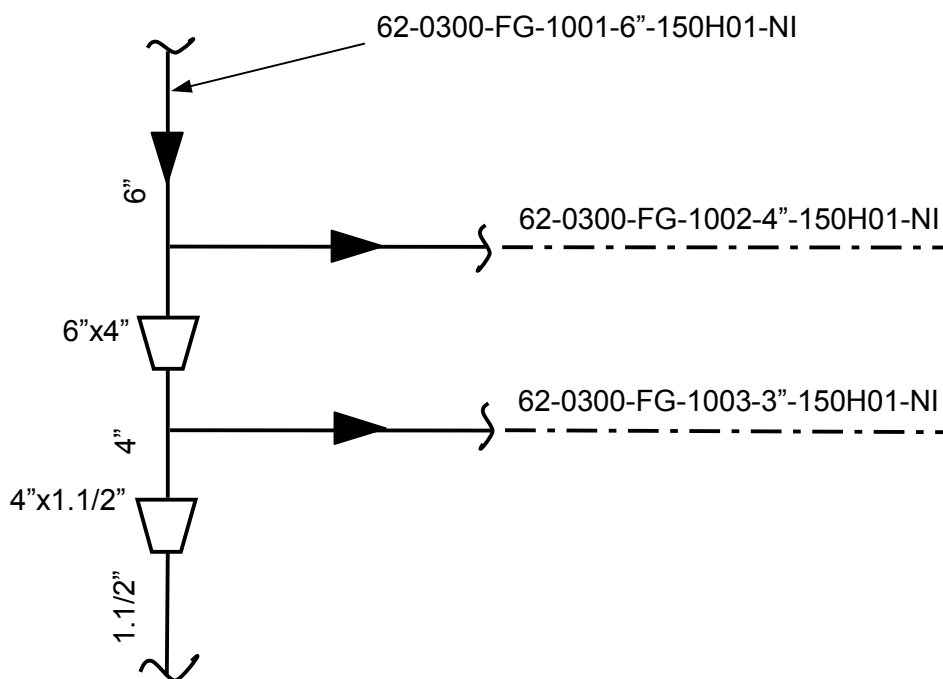
5.4 Process Engineering Group.

- (a) Initial draft of IP&ID containing new lines with Area and Unit code, fluid code, line size and pressure rating.
- (b) Providing critical line conditions, to include operating temperatures and pressures, design temperatures and pressures, fluid phase, insulation type. All other process line list information, including steam out condition.

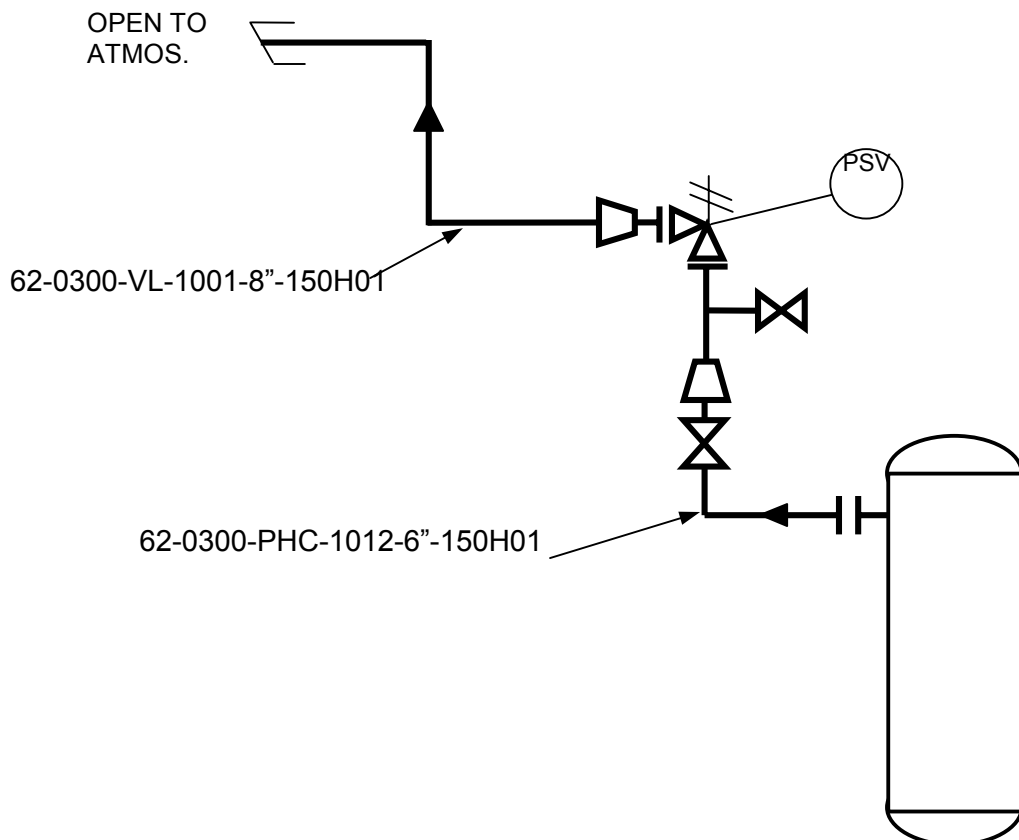
6. GENERAL RULES FOR LINE NUMBERING.

The following rules shall be used in assigning sequence numbers and in showing line designations on the IP&ID's.

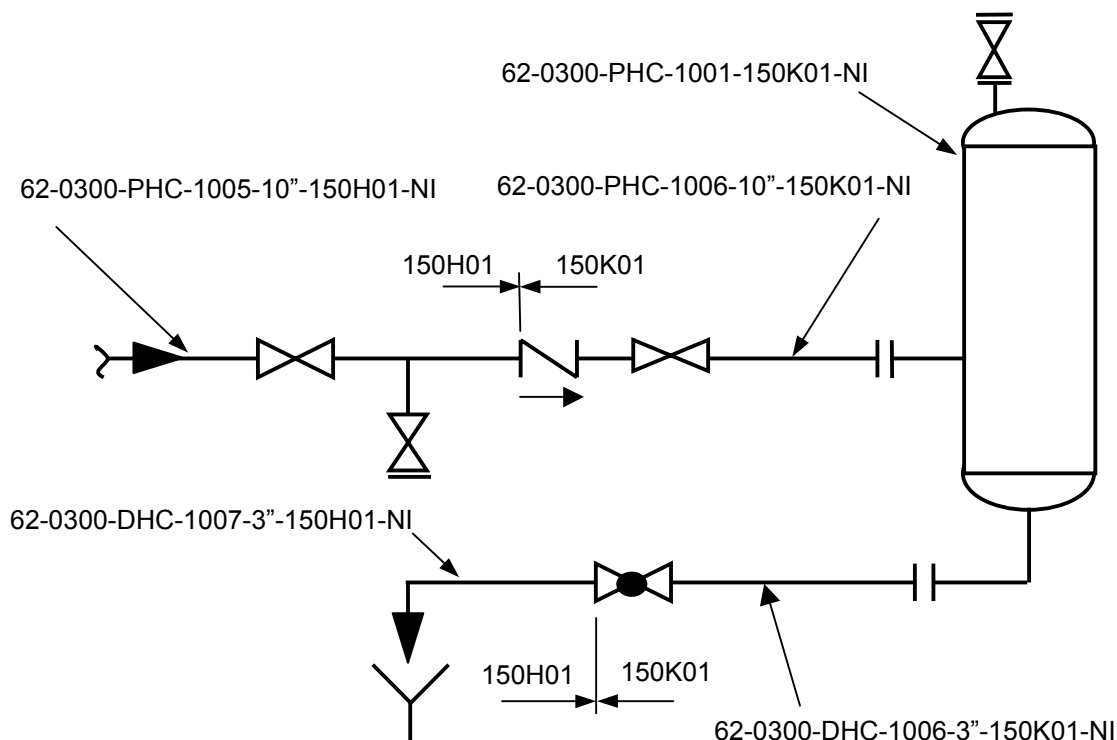
- 6.1** Use one sequence number for a line that runs from one piece of equipment to another, (except as indicated in para 6.7), in the direction of flow. The sequence number may include one or more line sizes. When a change in size occurs, a reducer shall be used to identify the other size on the IP&ID, (See example below). Short segments of pipe (i.e. control valve bypass, relief valve bypass) can be excluded from the line summary. However, bypass size will be identified on the IP&ID. The line list will report multiple entries for different pipe sizes.



- 6.2** Main process streams shall be numbered in process flow sequence where possible. Use the first utility system line sequence number for the utility header.
- 6.3** Sample connections and other short connections on a line shall not be assigned new numbers.
- 6.4** Short run inlet lines to relief valves shall have the same line sequence number as the line from which it originates. Longer runs and manifolded inlet lines to relief valves, shall have a different line sequence number from that of the line from which it originates.
- 6.5** Inlet lines to a relief valve from equipment shall have a different line sequence number from that of the equipment trim sequence number.
- 6.6** All lines from relief valves to collection headers shall be assigned different line sequence numbers from that of the collection header. All lines from relief valves, shall in all cases, be assigned a different line sequence number from that of the inlet line sequence number. Unique line sequence numbers shall be assigned for each relief valve outlet.

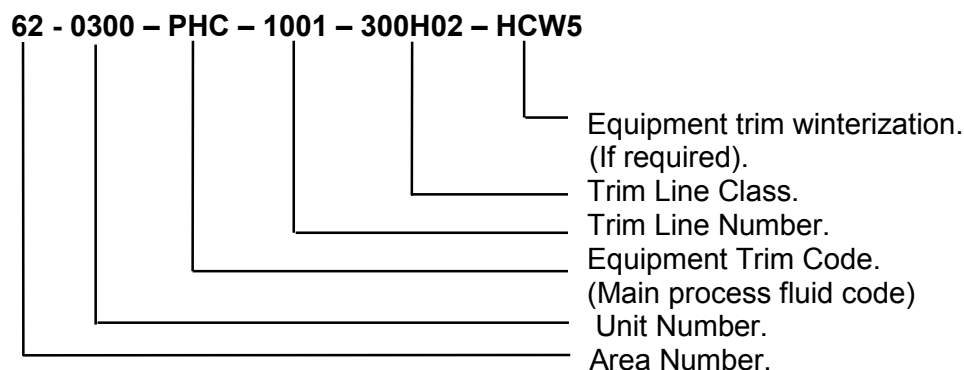


- 6.7** New line sequence numbers shall be assigned whenever the pipe class changes along the length of the line. Where the piping class break causes a change in pressure rating, the break point, wherever possible, will be at a flange connection. Specification breaks of this nature will not be at weld points. See example below.



- 6.8** The Vessel or Equipment trim shall be used to identify auxiliary connections around the vessel or equipment. The most stringent line class shall be used to identify these components. This includes vents, drains, instrument connections etc. When one line class cannot be used, a class break shall be used to clarify the difference. In this case, two vessel or equipment trim designators shall be shown on the IP&ID.

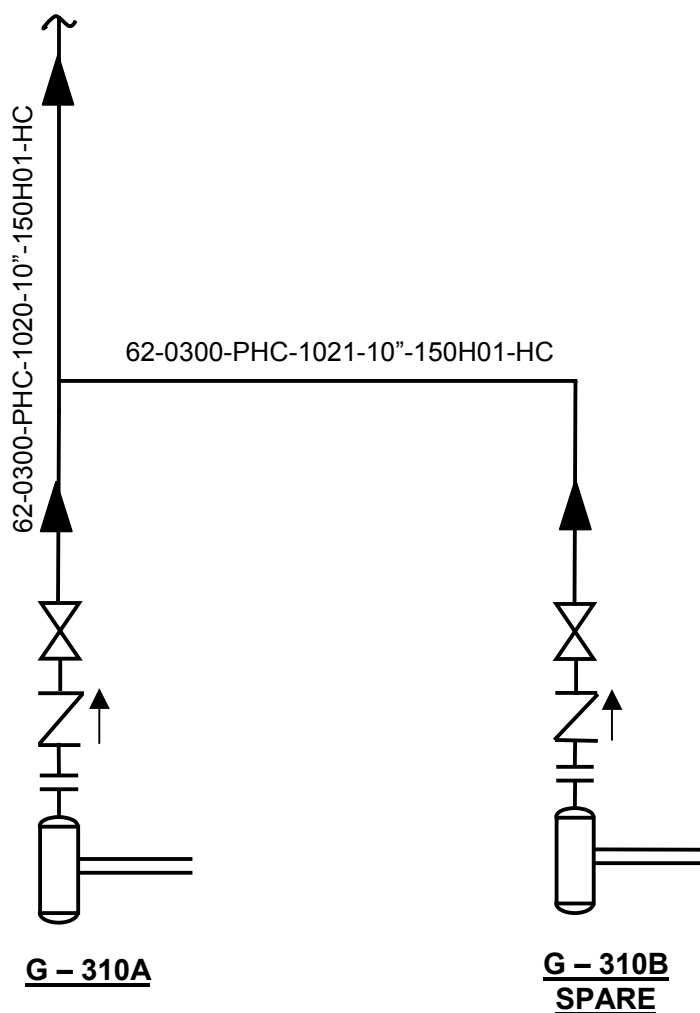
Equipment trim designators shall be shown as follows: -



Note:

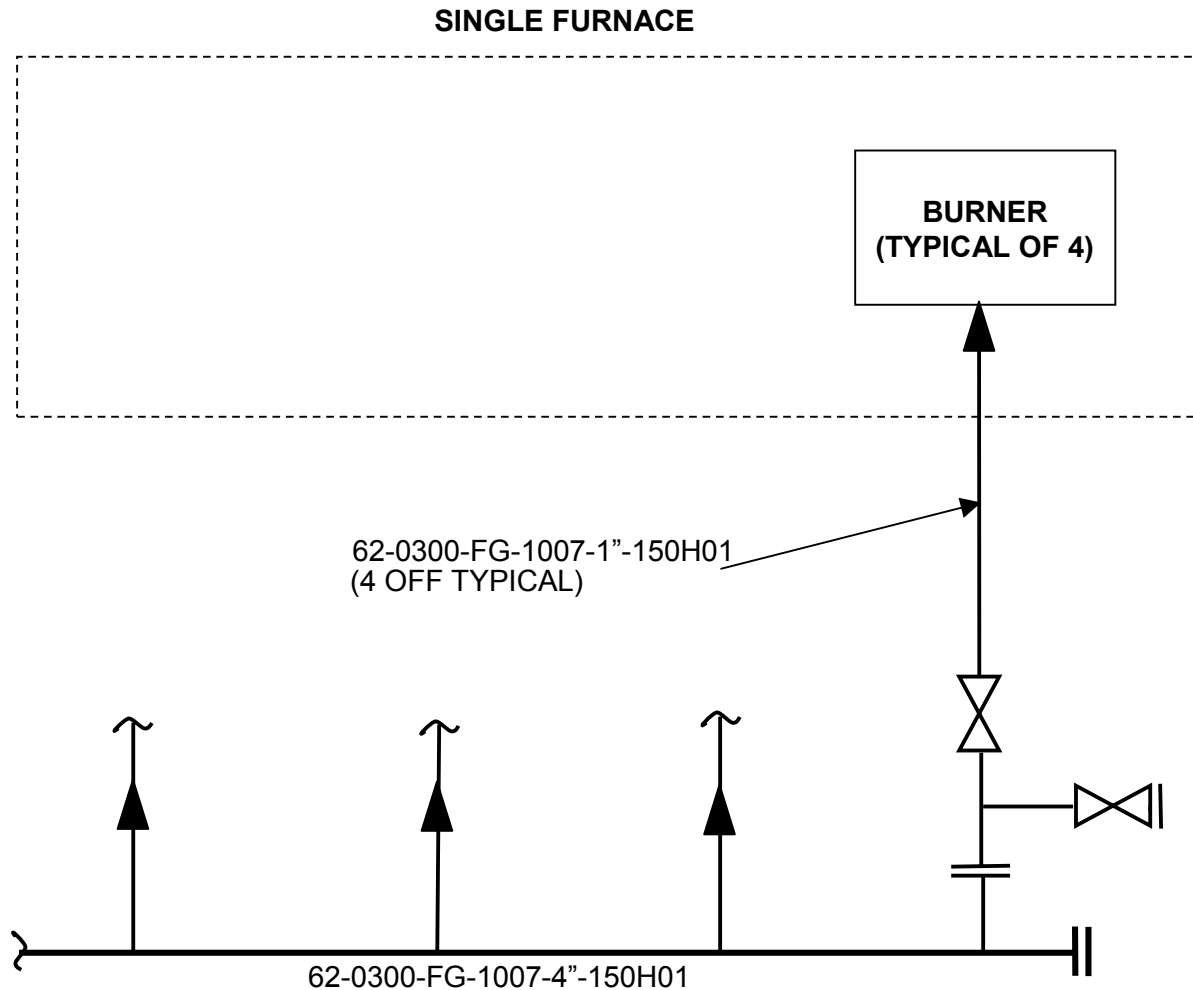
Equipment trim winterization is applicable for the trim only, equipment insulation shall be defined under the equipment label.

- 6.9** Where a line to or from a single source splits to multiple destinations, for example where equipment has both an operating and standby (spare) items (e.g. pumps or filters), a different line sequence number shall be assigned to the non-operating line. (See example below). When a pump is used as a common spare for two different services, a new line sequence number shall be assigned for the second service, as above. The common equipment shall carry the piping class of the more severe service.



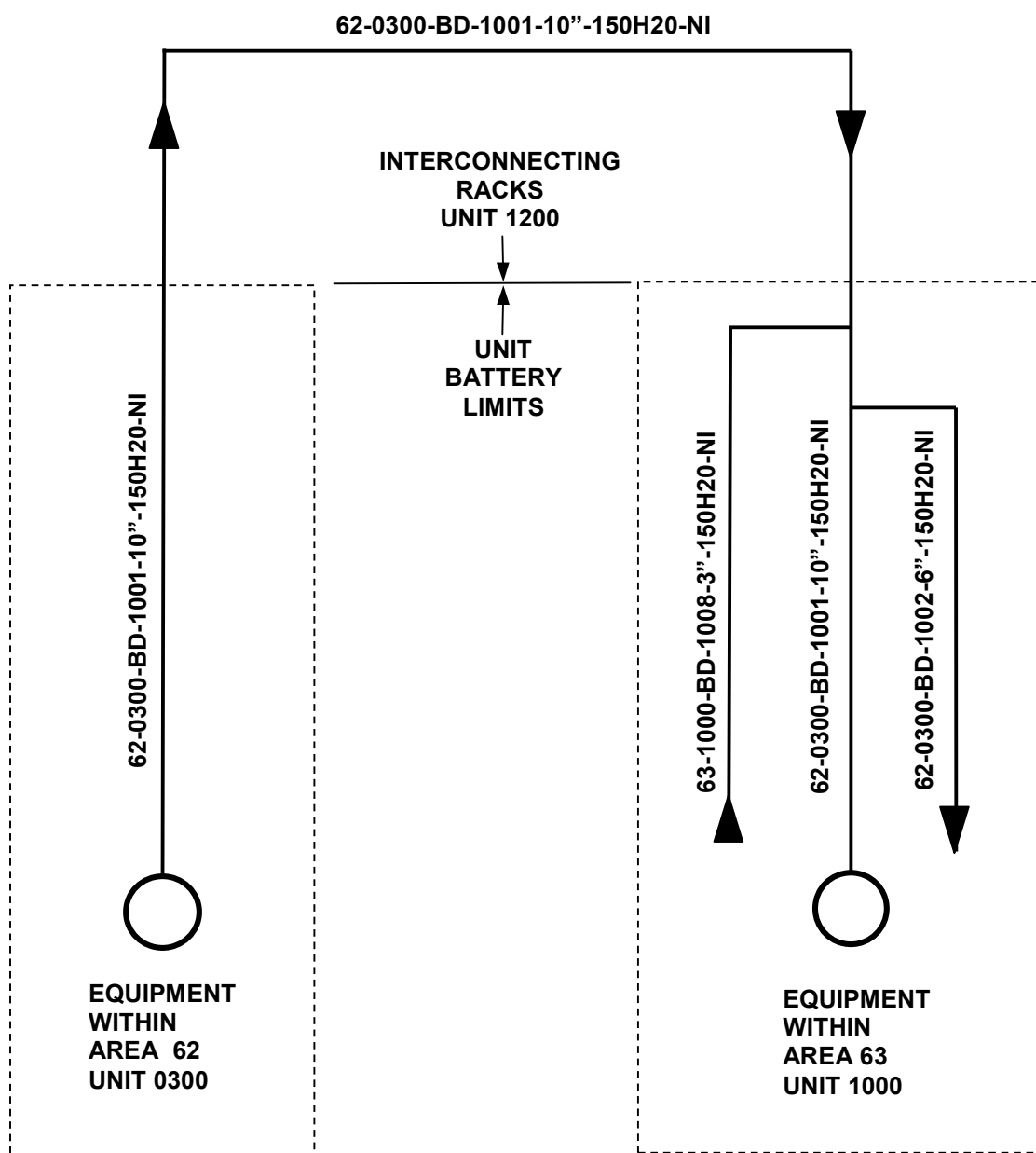
- 6.10** A line class break is not required at vessel/equipment nozzles.

- 6.11** Use only one line designator to identify header and manifolds for air coolers, burners etc. The line designator needs to be shown only once, but with a note to indicate how many typical identical manifolds there are. Branch sizes shall be identified on the IP&ID. The inlet and outlet piping shall each have a unique identifier. (See example below).

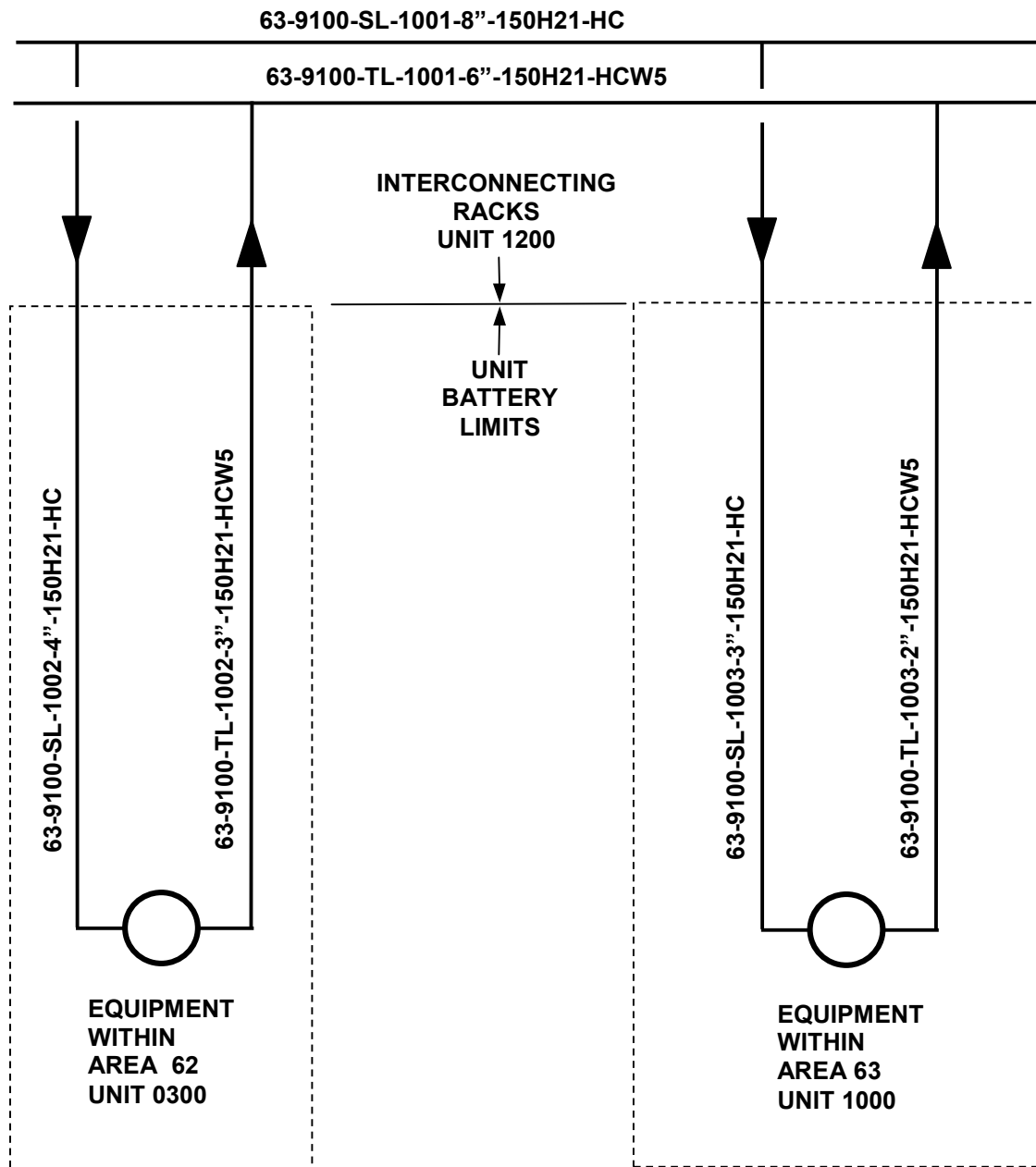


- 6.12** Where the insulation designation changes along a line, no new line sequence number is required, if the line conditions remain constant.
- 6.13** A deleted line sequence number or designator will not be re-used.

- 6.14** For a process line passing through a number of units, the line shall carry the complete number of the unit from which it originates. If the line serves multiple units, each branch shall carry the Area/Unit number of the originating line. (See example below).



- 6.15** For utility lines passing through a number of units, the line shall carry the complete number of the unit from which it originates. If the line serves multiple units, each branch shall carry the line number (with modified sequence number) of the originating line. (See example below).



7. PROCEDURES FOR LINE LIST GENERATION

Piping Engineering shall be responsible for line numbering on the project. The IP&ID Production Group shall enter the line numbers into the IP&ID's. A Piping Materials Engineer shall work within the Group to facilitate this activity. The Piping Materials Engineer shall be fully trained in the use of the IP&ID system. This method of working being consistent with the Piping ownership of the project line list.

The line list or summary on the project will be produced directly from the IP&ID database and issued to match each revision of the P&ID's from Rev.'B' onwards. Line list information will be a critical item to support OPTIMEYES pipe routing and downstream estimating activities/programmes. The following method of inputting the line number shall be adopted.

- 7.1** Initial drafts of the IP&ID shall contain the line number label, including Area and Unit code, against each new line. The process engineering group shall provide the Fluid code, Size, Pressure Rating and Insulation type in a format readily transferable to the IP&ID database. The IP&ID Production Group will then add the line sequence numbers, prior to the Rev.'B' issue of the document. The Piping Materials Engineer shall ensure the correct line numbering procedure is followed.
- 7.2** Process engineering group shall indicate where line conditions change. Other process line list information, such as steam out conditions, and whether Sour Service or not, shall be added by PE during the process of entering line conditions into the IP&ID database.
- 7.3** Special requirements, and other information, such as hydro-test pressure, NDT, PWHT, or painting code as required for piping fabrication and erection, shall be added into the IP&ID database via specified routines.
- 7.4** Each draft of the IP&ID shall be propagated to the database, and all errors addressed to ensure "OK" status, finally the drawing will be checked for completeness. A line summary for each P&ID shall accompany the check print for Process to check during the production cycle between Rev.'A' and Rev.'B'.
- 7.5** When all the steps above have been successfully completed, the Piping Materials Engineer can produce the line list or summary, directly from the IP&ID database in the correct project format.
- 7.6** Once all the data for an area is entered into the database a unit wide line list is produced and checked by Piping Material Engineering and with final input from Process. Updates are feed back into the database, including information generated by the routines.

7.7 When the line list is complete the data is “posted” to the master database for other disciplines to utilise.

7.8 A Flow Chart of the process is shown below.

